Tableau Big Data Visualization Blueprint

Role: Member 2 - Weather & Spatial Analytics Lead (Yashmitha) | Collaborator: Akilan Pon Ilango (Member 4 - ML)
Repository: https://github.com/therealakilan220/BeyondTheWickets.git

1. Quick Synchronization & Setup

Run in terminal: git pull origin main. Open **Tableau Public** (free download) or **Tableau Student**.
Connect to CSV file: **outputs/bowler_spells_dataset.csv** (or **SpaTemp Datasets/Arvind-fused dataset/fused_ipl_weather_pitch_ml.csv**).

2. Tableau Calculated Fields to Add

- **Dew Risk Zone:** `IF [dew_spread] <= 6.0 THEN "Severe Dew Risk" ELSEIF [dew_spread] <= 10.0 THEN "Moderate Dew" ELSE "Dry" END`
- **Dew Condensation Factor:** `[temperature] - [dew_point]`

3. Master Dashboard Sheets (Step-by-Step)

Sheet Name	Columns & Rows	Marks & Filters	Viva / Paper Narrative
Sheet 1: Stadium Dew Map	Cols: Longitude Rows: Latitude	Detail: venue Color: Dew Risk Zone Size: AVG(economy_rate)	Shows coastal stadiums (Chennai, Mumbai) have severe dew saturation.
Sheet 2: Chepauk vs Chinnaswamy	Cols: venue (filter to both) Rows: Humidity, Dew Spread, Wind Speed	Mark: Bar chart Color: venue	Contrasts Chennai coastal humidity (72.9%) vs Bengaluru high-altitude wind.
Sheet 3: 2nd Innings Grip Loss Heatmap	Cols: Dew Risk Zone Rows: bowler_type	Mark: Square / Heatmap Color & Label: AVG(economy_rate) Filter: innings = 2	Proves pace economy rises +0.86 RPO and spin +1.14 RPO under severe dew.
Sheet 4: Climate Trend (2008-2026)	Cols: season_year Rows: AVG(temperature), AVG(dew_point)	Mark: Line chart (Dual Axis)	Proves sensor data integrity and zero missing values across 18 IPL seasons.

4. Dashboard Assembly & Interactive Filter

Combine Sheet 1 (Top 50%) with Sheet 2 and Sheet 3 (Bottom 50%). Click the filter icon on Sheet 1 (**"Use as Filter"**) so clicking any venue on the map immediately filters the charts below!

5. Exporting for Viva & Paper

- **Viva Demo:** File -> Save to Tableau Public As... (provides an interactive web URL for browser demo).
- **Paper:** Dashboard -> Export Image (300 DPI vector graphic for the IEEE manuscript).